## Investigation: OSX BRASIL S.A. (CVM code 21342), fiscal year 2012, as it was on 2013-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

## Company

Sector: Emp. Adm. Part. - Sem Setor Principal (origin: FCA); group: non-financial.
nenhum código de negociação no FCA entregue até a data (nos dados abertos, o campo só vem preenchido a partir do FCA de 2018)

## Document used

2012 DFP, version 2, filed on 2013-04-05; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2013-02-27); 2 (2013-04-05).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=25967

## Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: −1.091 (index)
- Alert by the study 1 rule: yes
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: 89 (n = 22)
- Percentile in group and year: 92 (n = 334)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 0.885 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 41 (n = 23)
- Percentile in group and year: 27 (n = 412)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

GMI, gross margin index
- Value: 1.087 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 78 (n = 25)
- Percentile in group and year: 68 (n = 426)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

AQI, asset quality index
- Value: 0.947 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 33 (n = 26)
- Percentile in group and year: 27 (n = 421)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

SGI, sales growth index
- Value: 3.007 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 98 (n = 25)

Percentile in group and year: 98 (n = 427)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: 0.621 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 2 (n = 23)
Percentile in group and year: 8 (n = 351)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 0.342 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 2 (n = 25)
Percentile in group and year: 2 (n = 420)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.036 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 54 (n = 27)
Percentile in group and year: 55 (n = 438)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 1.684 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 94 (n = 27)
Percentile in group and year: 97 (n = 432)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: −865.1 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 27.2 days)
Percentile in sector and year: 2 (n = 22)
Percentile in group and year: 1 (n = 396)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: −6.36 p.p.
Alert by the study 1 rule: no
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 10 (n = 24)
Percentile in group and year: 3 (n = 394)
The company's own history: fewer than 5 previous fiscal years with a value

Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.049 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 43 (n = 27)
Percentile in group and year: 49 (n = 437)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

## Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: −0.111 (index)
Components: X1 = −0.089; X2 = −0.013; X3 = −0.004; X4 = 0.515 (indices)
Percentile in sector and year: 36 (n = 66)
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

## Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 3/9 (8 computable)
Signals:
- Return on assets (ROA): 0
- Operating cash flow on assets (CFO): 1
- Change in ROA: 0
- Cash above earnings (ACCRUAL): 1
- Change in leverage: 0
- Change in current ratio: 0
- No equity issuance (EQ_OFFER): —
- Change in gross margin: 0
- Change in asset turnover: 1
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

## Changes

Restatement through the following year's comparative: no difference above the tolerance, or the next year's DFP did not exist yet on the date.

## What was known on the date

- DFP 2010, v1: 2011-03-15
- DFP 2010, v2: 2011-03-15
- FCA 2012, v1: 2012-01-26
- DFP 2011, v1: 2012-03-20
- DFP 2011, v2: 2012-03-30
- FCA 2012, v2: 2012-05-31
- FCA 2012, v3: 2012-06-26
- DFP 2012, v1: 2013-02-27

- DFP 2012, v2: 2013-04-05
DFP and FCA filed with the CVM by the reference date; ITR comes in a later phase.
Material facts and market notices filed by the date (IPE open data, since 2010): 23 material fact documents and 63 market notice documents.
Material facts filed in the 24 months before the date (from 2011-05-01 to 2013-04-30):
- 2012-10-17: subject omitted (possible person's name); see the document at the CVM
- 2012-08-13: subject omitted (possible person's name); see the document at the CVM
- 2012-06-28: OSX anuncia novo Diretor Presidente
- 2012-06-15: subject omitted (possible person's name); see the document at the CVM
- 2012-04-27: subject omitted (possible person's name); see the document at the CVM
- 2012-04-19: subject omitted (possible person's name); see the document at the CVM
- 2012-03-14: subject omitted (possible person's name); see the document at the CVM
- 2011-10-31: Desdobramento das ações ordinárias
- 2011-06-27: subject omitted (possible person's name); see the document at the CVM
Older material facts, the market notices and the link to each document are in the JSON and on screen.
Material facts and market notices filed with the CVM by the date, from the IPE open data (since 2010): filing date, category, subject (as filed, in Portuguese) and link. The content of the documents is not downloaded, read or summarized, and the list generates no alert, score or ranking. The counts are of documents: versions, resubmissions and English versions count separately; in the list, versions of the same document and documents with the same date, category and subject appear on a single line, with a link to each one.

## Data quality

53 applicable checks; 0 failed.

## Auditor's report and auditor (factual section; no alert)

Auditor's report: Sem Ressalva (unqualified).
Excerpt checked literally: "Em nossa opinião, as demonstrações financeiras individuais acima referidas apresentam adequadamente, em todos os aspectos relevantes, a posição patrimonial e financeira da OSX Brasil S.A em 31 de dezembro de 2012, o desempenho de suas operações e os seus fluxos de caixa para o exercício findo naquela data, de acordo com as práticas contábeis adotadas no Brasil."
Auditor in the 2010 FCA: KPMG Auditores Independentes.
Auditor in the 2011 FCA: KPMG Auditores Independentes.
Auditor in the 2012 FCA: Ernst & Young Terco Auditores Independentes S/S.
Auditor changes (one line per change, canonical name by the firm's CVM code):
- 2012: KPMG Auditores Independentes -> Ernst & Young Terco Auditores Independentes S/S. The outgoing firm audited 3 consecutive fiscal years (2009 to 2011), by the start and end dates in the FCA.

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

## Document excerpts (copied; no alert)

Excerpts copied from the documents, with the page, and the Reference Form table. It is not a signal family: it generates no alert, score or ranking. Absence of mention does not prove absence.
How the reader was tested: 20 held-out cases, a single measurement with the frozen reader (c99c953): precision 0.962, coverage 0.962, literal check on 132 of 132 excerpts, no natural-person name (docs/leitor_medida.json).

Explanatory notes · Related parties

Note: 20. Partes relacionadas (pages 44 to 53)

- [subsidiaries] “operações com partes relacionadas, bem como as transações que influenciaram o resultado do período, são decorrentes de transações da Companhia com empresas controladas e ligadas, conforme se descreve a seguir.” (p. 44, original excerpt in Portuguese)
- Excerpt omitted; see page 45 of the document
- [other related parties] “a) Empresas ligadas” (p. 45, original excerpt in Portuguese)
- [subsidiaries; other related parties] “A Companhia possui como principais empresas ligadas, com as quais mantém alguma transação, as seguintes: EBX Participações Ltda., EBX Investimentos Ltda., EBX Holding Ltda., OGX Petróleo e Gás Participações Ltda., LLX Logística S.A. (“LLX”), MPX Energia S.A. (“MPX”), MMX Mineração e Metálicos S.A. e suas controladas (“MMX”) e AVX Táxi Aéreo Ltda. (“AVX”).” (p. 45, original excerpt in Portuguese)
- [parent company] “Controladora Contas a receber Contas a pagar Resultado 31/12/2012 31/12/2011 31/12/2012 31/12/2011 31/12/2012 31/12/2011” (p. 45, original excerpt in Portuguese)
- [subsidiaries; other related parties] “relativos a operações com partes relacionadas, decorrem de transações da Companhia com suas controladas diretas e indiretas e com empresas ligadas, as quais foram realizadas em condições usuais de mercado para os respectivos tipos de operações, conforme descrito abaixo:” (p. 45, original excerpt in Portuguese)
- [subsidiaries] “(i) Refere-se ao contrato de compartilhamento de recursos e serviços administrativos firmados entre a OSX Brasil e suas controladas, a EBX Investimentos Ltda. e a EBX Holding Ltda., por meio do qual estas empresas executam serviços de administração de caixa e administração financeira, serviços legais, seguros e auditorias internas, governança corporativa, comunicações, compras e recursos humanos, entre outros. Mensalmente a EBX efetua a medição e calcula o valor das atividades sujeitas aos custos compartilhados efetivamente” (p. 45, original excerpt in Portuguese)
- Excerpt omitted; see page 46 of the document
- [other related parties] “e até 23 de março de 2013, a OSX Brasil possa exigir que tais acionistas” (p. 46, original excerpt in Portuguese)
- [parent company] “controladores subscrevam novas ações até o limite máximo de USD 1,0 bilhão, ao preço por ação equivalente ao praticado na oferta pública inicial de ações da OSX Brasil, corrigido conforme a variação do IGP-M, por meio de aumento de capital privado a ser realizado em conformidade com os artigos 170 e seguintes da Lei das Sociedades por Ações. A opção poderá ser exercida na hipótese de se verificar a necessidade de capital adicional pela OSX Brasil para a realização de seu plano de negócios e a ausência de alternativas para tal captação junto aos mercados.” (p. 47, original excerpt in Portuguese)
- Excerpt omitted; see page 50 of the document
- [subsidiaries] “em 13 de dezembro de 2012, a subsidiária OSX Serviços Operacionais Ltda., e” (p. 50, original excerpt in Portuguese)
- [key management personnel] “20.4. Remuneração dos membros do Conselho de Administração e da Diretoria” (p. 51, original excerpt in Portuguese)
- [key management personnel; other related parties] “De acordo com a Lei nº 6.404/1976 e com o estatuto social da Companhia, é responsabilidade dos acionistas, em Assembleia Geral, fixar o montante global da remuneração anual dos administradores. Cabe ao Conselho de Administração efetuar a distribuição da verba entre os administradores.” (p. 51, original excerpt in Portuguese)
- [key management personnel] “Desta forma, os montantes referentes à remuneração anual da Diretoria, Comitê de Auditoria e Conselho de Administração estão apresentados abaixo:” (p. 52, original excerpt in Portuguese)
- Excerpt omitted; see page 52 of the document
- [key management personnel] “Opções de ações outorgadas (**) 4.506 41.089 4.506 41.089” (p. 52, original excerpt in Portuguese)
- [key management personnel] “9.532 46.475 9.532 46.475” (p. 52, original excerpt in Portuguese)

Fields of item 18 of CPC 05 (R1)

- [balances] “Os principais saldos de ativos e passivos em 31 de dezembro de 2012, relativos a” (p. 44, original excerpt in Portuguese)
- [balances] “Controladora Contas a receber Contas a pagar Resultado 31/12/2012 31/12/2011

31/12/2012 31/12/2011 31/12/2012 31/12/2011" (p. 45, original excerpt in Portuguese)
- [balances] "Consolidado Contas a pagar/ Contas a receber adiantamento de clientes Capex Cessão onerosa Resultado 31/12/2012 31/12/2011 31/12/2012 31/12/2011 31/12/2012 31/12/2011 31/12/2012 31/12/2011" (p. 45, original excerpt in Portuguese)
- [balances] "Os principais saldos de ativos e passivos em 31 de dezembro de 2012," (p. 45, original excerpt in Portuguese)
- [amount of the transactions] "(i) Refere-se ao contrato de compartilhamento de recursos e serviços administrativos firmados entre a OSX Brasil e suas controladas, a EBX Investimentos Ltda. e a EBX Holding Ltda., por meio do qual estas empresas executam serviços de administração de caixa e administração financeira, serviços legais, seguros e auditorias internas, governança corporativa, comunicações, compras e recursos humanos, entre outros. Mensalmente a EBX efetua a medição e calcula o valor das atividades sujeitas aos custos compartilhados efetivamente" (p. 45, original excerpt in Portuguese)
- [amount of the transactions; balances] "na rubrica de despesas gerais e administrativas, o montante de R$ 34.515, referente ao contrato acima referido. O saldo a pagar deste" (p. 46, original excerpt in Portuguese)
- Excerpt omitted; see page 46 of the document
- [terms and conditions] "controladores subscrevam novas ações até o limite máximo de USD 1,0 bilhão, ao preço por ação equivalente ao praticado na oferta pública inicial de ações da OSX Brasil, corrigido conforme a variação do IGP-M, por meio de aumento de capital privado a ser realizado em conformidade com os artigos 170 e seguintes da Lei das Sociedades por Ações. A opção poderá ser exercida na hipótese de se verificar a necessidade de capital adicional pela OSX Brasil para a realização de seu plano de negócios e a ausência de alternativas para tal captação junto aos mercados." (p. 47, original excerpt in Portuguese)
- Excerpt omitted; see page 47 of the document
- [terms and conditions] "valor de USD 188,3 milhões, a uma taxa de juros de Libor + 2,90% a.a.. Em" (p. 47, original excerpt in Portuguese)
- [terms and conditions] "valor de USD 55,0 milhões, a uma taxa de juros de Libor + 2,90% a.a.. Em 07" (p. 47, original excerpt in Portuguese)
- [terms and conditions] "milhões de principal e USD 655 mil de juros. Em 07 de março de 2012, ocorreu" (p. 47, original excerpt in Portuguese)
- [terms and conditions] "15,3 milhões de principal e USD 142 mil de juros." (p. 47, original excerpt in Portuguese)
- [terms and conditions] "valor de USD 8,0 milhões, a uma taxa de juros de Libor + 2,90% a.a.. Em maio" (p. 47, original excerpt in Portuguese)
- [terms and conditions] "mil de juros." (p. 47, original excerpt in Portuguese)
- [terms and conditions] "valor de USD 7,0 milhões, a uma taxa de juros de Libor + 2,90% a.a.. Em maio" (p. 47, original excerpt in Portuguese)
- [terms and conditions] "valor de USD 168,2 milhões, a uma taxa de juros de Libor + 2,90% a.a.. Em" (p. 47, original excerpt in Portuguese)
- [terms and conditions] "USD 1,9 milhões de juros." (p. 47, original excerpt in Portuguese)
- [terms and conditions] "valor de USD 11,6 milhões, a uma taxa de juros de Libor + 2,90% a.a..Em" (p. 48, original excerpt in Portuguese)
- [terms and conditions] "valor de USD 20,0 milhões, a uma taxa de juros de Libor + 2,90% a.a.." (p. 48, original excerpt in Portuguese)
- [terms and conditions] "no valor de USD 41.992 mil, a uma taxa de juros de Libor + 2,90% a.a.. Em" (p. 48, original excerpt in Portuguese)
- Excerpt omitted; see page 48 of the document
- [terms and conditions] "no valor total de USD 49,5 milhões, a uma taxa de juros de Libor + 2,90% a.a.." (p. 48, original excerpt in Portuguese)
- [terms and conditions] "de USD 250,0 milhões, a uma taxa de juros de Libor + 2,90% a.a.." (p. 48, original excerpt in Portuguese)
- [terms and conditions] "valor de USD 10,3 milhões, a uma taxa de juros de Libor + 2,90% a.a.." (p. 48, original excerpt in Portuguese)
- [terms and conditions] "milhões, respectivamente, a uma taxa de juros de Libor + 2,90% a.a.." (p. 48, original excerpt in Portuguese)

- [terms and conditions] “de USD 1.300 mil, a uma taxa de juros de Libor + 2,90% a.a.. Em setembro de” (p. 48, original excerpt in Portuguese)
- [terms and conditions] “juros.” (p. 48, original excerpt in Portuguese)
- [terms and conditions] “de USD 5.000 mil, a uma taxa de juros de Libor + 2,90% a.a.. Em setembro de” (p. 48, original excerpt in Portuguese)
- [terms and conditions] “uma taxa de juros de Libor + 2,90% a.a..” (p. 48, original excerpt in Portuguese)
- [terms and conditions] “valor de USD 13.500 mil, a uma taxa de juros de Libor + 2,90% a.a..” (p. 48, original excerpt in Portuguese)
- [terms and conditions] “valor de USD 10.000 mil, a uma taxa de juros de Libor + 2,90% a.a..” (p. 48, original excerpt in Portuguese)
- [terms and conditions] “valor de USD 19.900 mil, a uma taxa de juros de Libor + 2,90% a.a..” (p. 49, original excerpt in Portuguese)
- [terms and conditions] “no valor de USD 19.200 mil, a uma taxa de juros de Libor + 2,90% a.a..” (p. 49, original excerpt in Portuguese)
- [terms and conditions] “valor de USD 900 mil, a uma taxa de juros de Libor + 2,90% a.a.. Em dezembro” (p. 49, original excerpt in Portuguese)
- [terms and conditions] “juros.” (p. 49, original excerpt in Portuguese)
- [terms and conditions] “no valor de USD 19.800 mil, a uma taxa de juros de Libor + 2,90% a.a..” (p. 49, original excerpt in Portuguese)
- [terms and conditions] “valor de USD 350 mil, a uma taxa de juros de Libor + 2,90% a.a..” (p. 49, original excerpt in Portuguese)
- [terms and conditions] “valor de USD 290 mil, a uma taxa de juros de Libor + 2,90% a.a..” (p. 49, original excerpt in Portuguese)
- [terms and conditions] “no valor de USD 32.100 mil, a taxa de juros de Libor + 2,90% a.a..” (p. 49, original excerpt in Portuguese)
- [terms and conditions] “no valor de USD 34.000 mil, a uma taxa de juros de Libor + 2,90% a.a..” (p. 49, original excerpt in Portuguese)
- [terms and conditions] “valor de USD 25.000 mil, a uma taxa de juros de Libor + 2,90% a.a..” (p. 49, original excerpt in Portuguese)
- [terms and conditions] “valor de USD 6.000 mil, a uma taxa de juros de Libor + 2,90% a.a..” (p. 49, original excerpt in Portuguese)
- [terms and conditions] “valor de USD 50.000 mil, a uma taxa de juros de Libor + 2,90% a.a..” (p. 49, original excerpt in Portuguese)
- [terms and conditions] “valor de USD 75.000 mil, a uma taxa de juros de Libor + 2,90% a.a..” (p. 49, original excerpt in Portuguese)
- [terms and conditions] “no valor de USD 5.900 mil, a uma taxa de juros de Libor + 2,90% a.a..” (p. 50, original excerpt in Portuguese)
- [terms and conditions] “no valor de USD 19.100 mil, a uma taxa de juros de Libor + 2,90% a.a..” (p. 50, original excerpt in Portuguese)
- Excerpt omitted; see page 50 of the document
- [guarantees] “20.3. Garantias” (p. 50, original excerpt in Portuguese)
- Excerpt omitted; see page 51 of the document
- [amount of the transactions] “De acordo com a Lei nº 6.404/1976 e com o estatuto social da Companhia, é responsabilidade dos acionistas, em Assembleia Geral, fixar o montante global da remuneração anual dos administradores. Cabe ao Conselho de Administração efetuar a distribuição da verba entre os administradores.” (p. 51, original excerpt in Portuguese)
- [amount of the transactions] “Desta forma, os montantes referentes à remuneração anual da Diretoria, Comitê de Auditoria e Conselho de Administração estão apresentados abaixo:” (p. 52, original excerpt in Portuguese)
- [amount of the transactions] “(**) As opções de compra de ações outorgadas estão des critas na Nota Explicativa nº 25.” (p. 52, original excerpt in Portuguese)

Market terms

- “relativos a operações com partes relacionadas, decorrem de transações da Companhia com suas controladas diretas e indiretas e com empresas ligadas, as quais foram realizadas em condições usuais de mercado para os respectivos tipos de operações, conforme descrito abaixo:” (p. 45,

original excerpt in Portuguese)
The sentence about market terms is recorded as a fact, without proof (CPC 05 (R1), item 23).

Explanatory notes · Reverse factoring and forfaiting
The reader found no mention of reverse factoring (risco sacado), forfaiting or supplier agreements in the notes of this DFP. Absence of mention does not prove absence of the operation.

Auditor's report
Type of opinion: unqualified
- [emphasis of matter] “Ênfase Conforme descrito na nota explicativa 2, as demonstrações financeiras individuais foram elaboradas de acordo com as práticas contábeis adotadas no Brasil. No caso da OSX Brasil S.A., essas práticas diferem do IFRS, aplicável às demonstrações financeiras separadas, somente no que se refere à avaliação dos investimentos em controladas, coligadas e controladas em conjunto pelo método de equivalência patrimonial, enquanto que para fins de IFRS seria custo ou valor justo. Nossa opinião não está ressalvada em função desse assunto. As demonstrações financeiras foram preparadas considerando a continuidade normal dos negócios da Companhia e de suas controladas, incluindo aquelas mencionadas na Nota Explicativa 4 e que se encontram em fase pré-operacional. A recuperação dos valores registrados no ativo não circulante depende do sucesso das operações futuras da Companhia e de suas controladas, bem como do suporte financeiro dos acionistas e/ou recursos de terceiros até que suas operações se tornem rentáveis. Os planos da administração com relação às atividades operacionais estão descritos também na Nota Explicativa 4.” (original excerpt in Portuguese)
- Excerpt omitted; see the auditor's report in the document

Reference Form
The Reference Form for this fiscal year was not available on the date. It is the only version available in the open data; earlier versions are not there.

Limits of this section:
- In real documents, about 22% of the excerpts come out omitted by the name detector, including names of companies outside the registry.
- In tables, header lines (years, column titles) may come out as excerpts; the held-out cases had no tables with multi-line headers.

## How the engine was tested
In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

## Limits
- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

## Limitations
- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 39 DFP lines, each with document, version, filing date, account and location.

## Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a F1); leitor-f1-1 (commit c99c953). The generation date is in the demo manifest (demo/manifesto.json).

## Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.
Name in today's registry: OSX BRASIL S.A. - EM RECUPERAÇÃO JUDICIAL.